

KOHLER AI Bathroom Designer

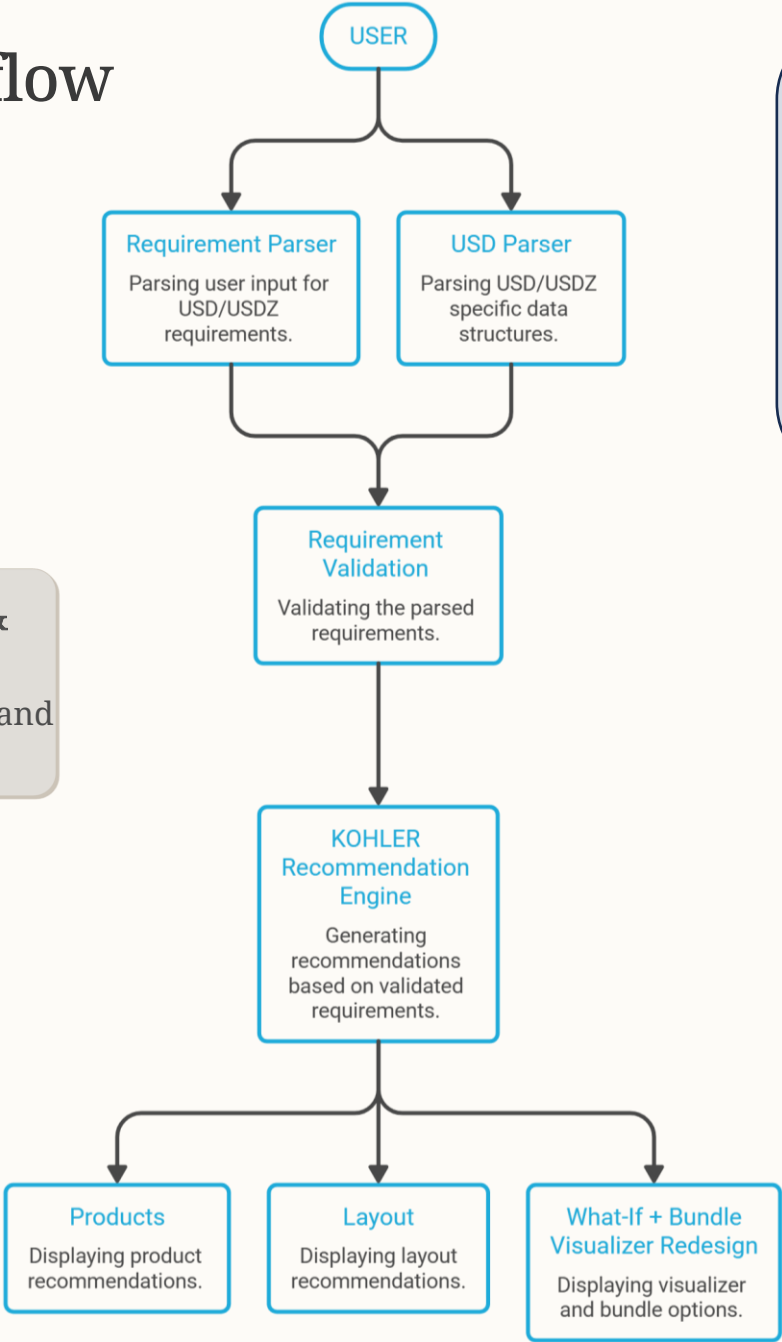
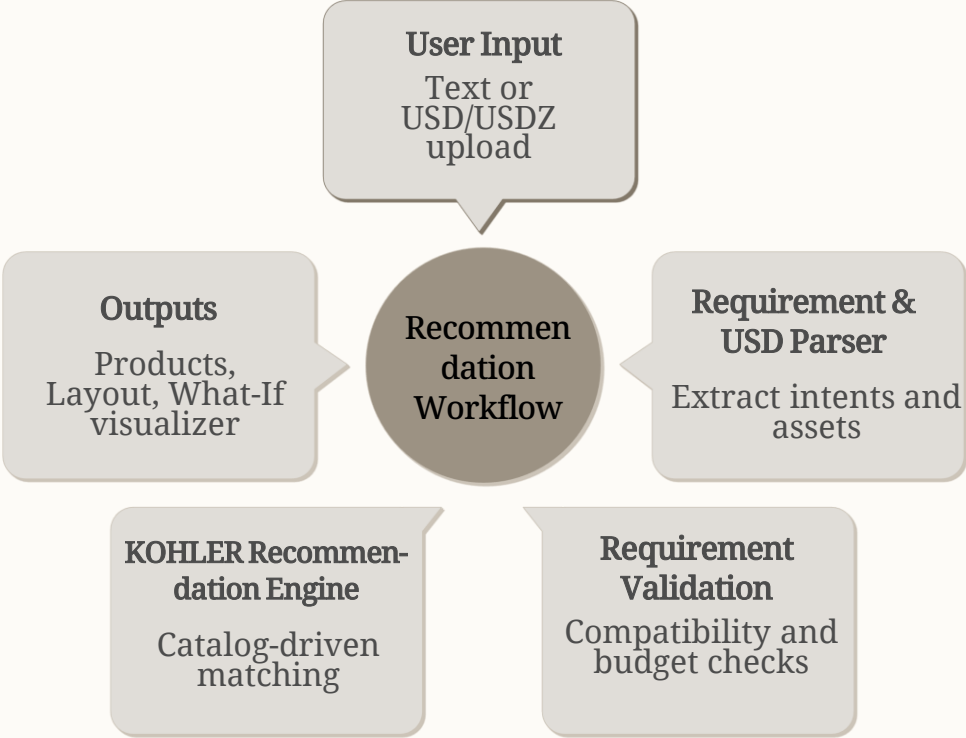
Bathroom planning requires combining **space dimensions**, user preferences, product compatibility, and budget — a multi-dimensional task that is slow and error-prone when done manually. Architectural files (USD / USDZ) contain rich spatial data but are rarely plug-and-play for product selection.

Our solution: an AI-assisted system that converts natural language requirements or USD/USDZ architecture into **validated, budget-aware KOHLER product recommendations** and a conceptual bathroom layout — streamlining planning from intent to purchase-ready bundles.

Key capability flows:

- Natural language → Requirement understanding → KOHLER catalog → Recommendations
- USD/USDZ → Spatial analysis → User product selection → Layout + Recommendations

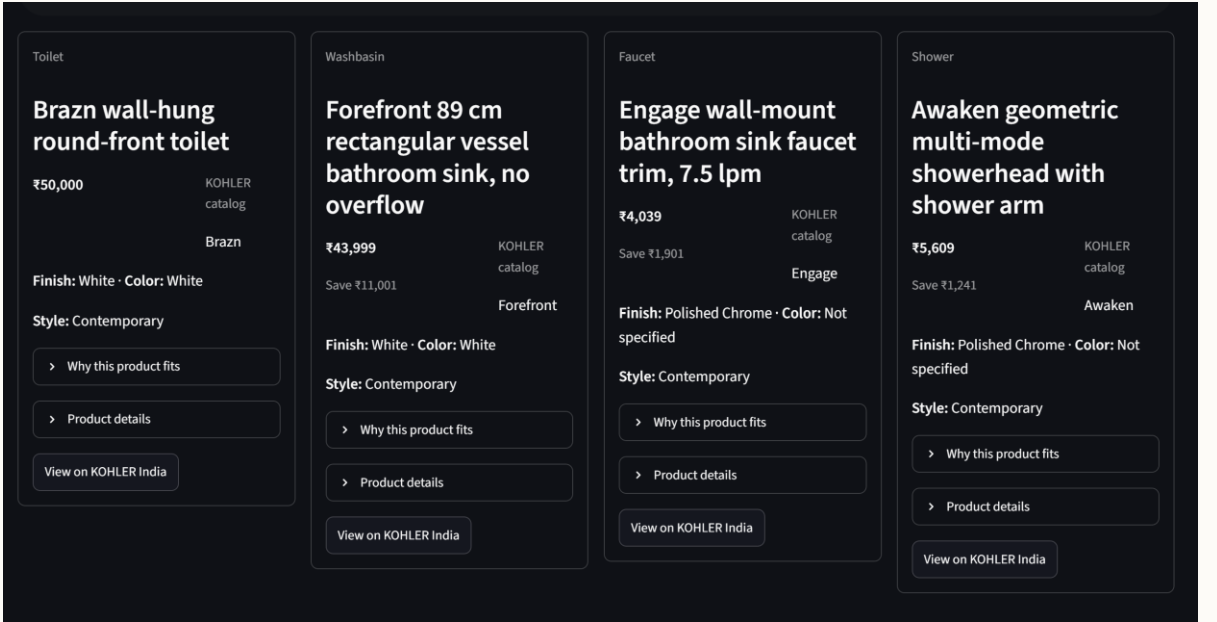
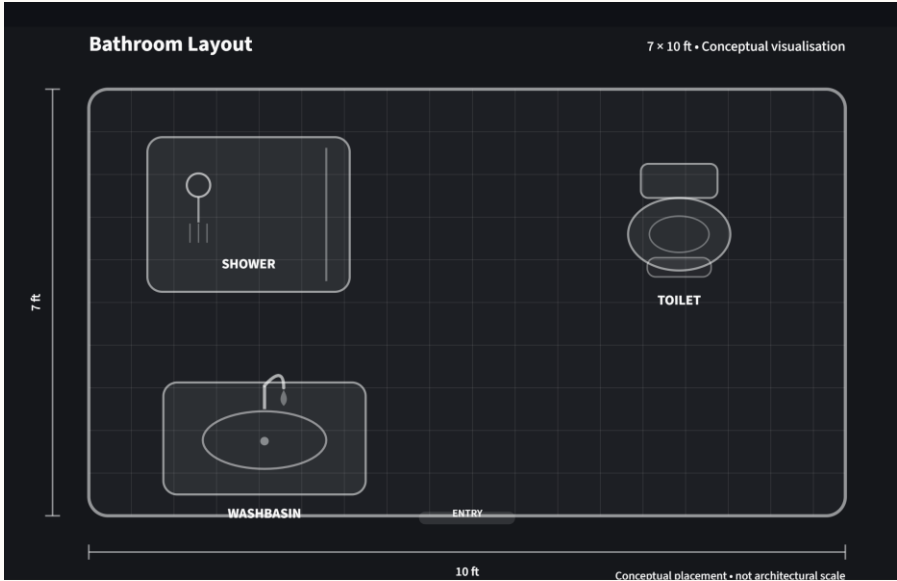
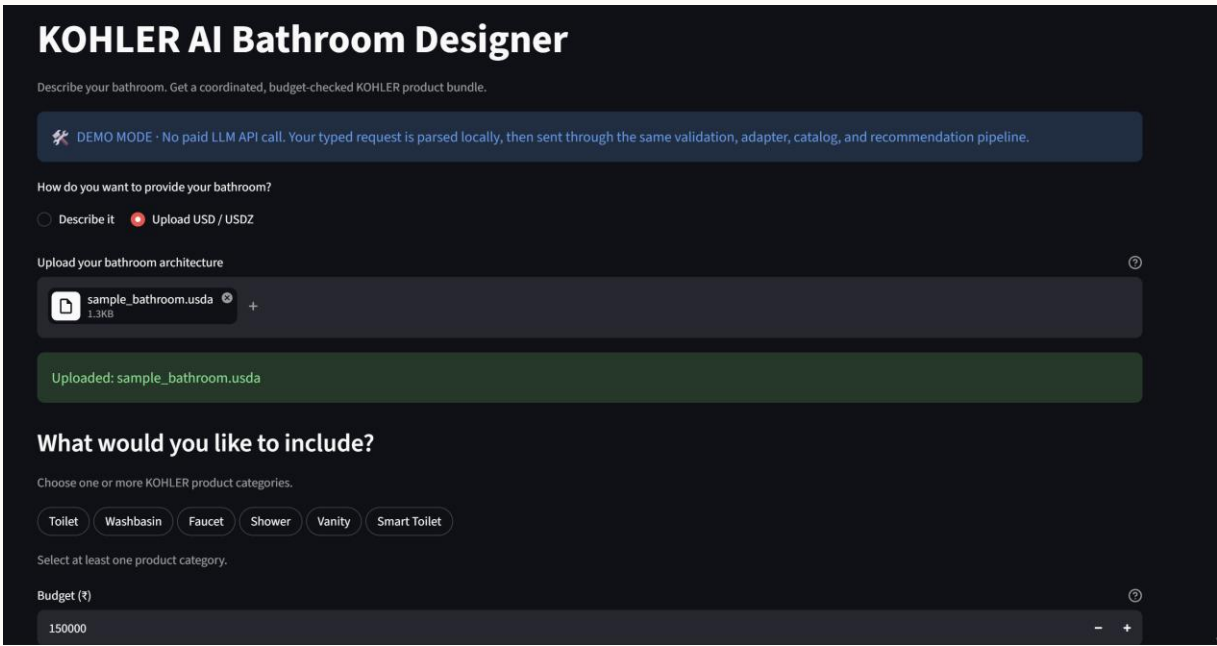
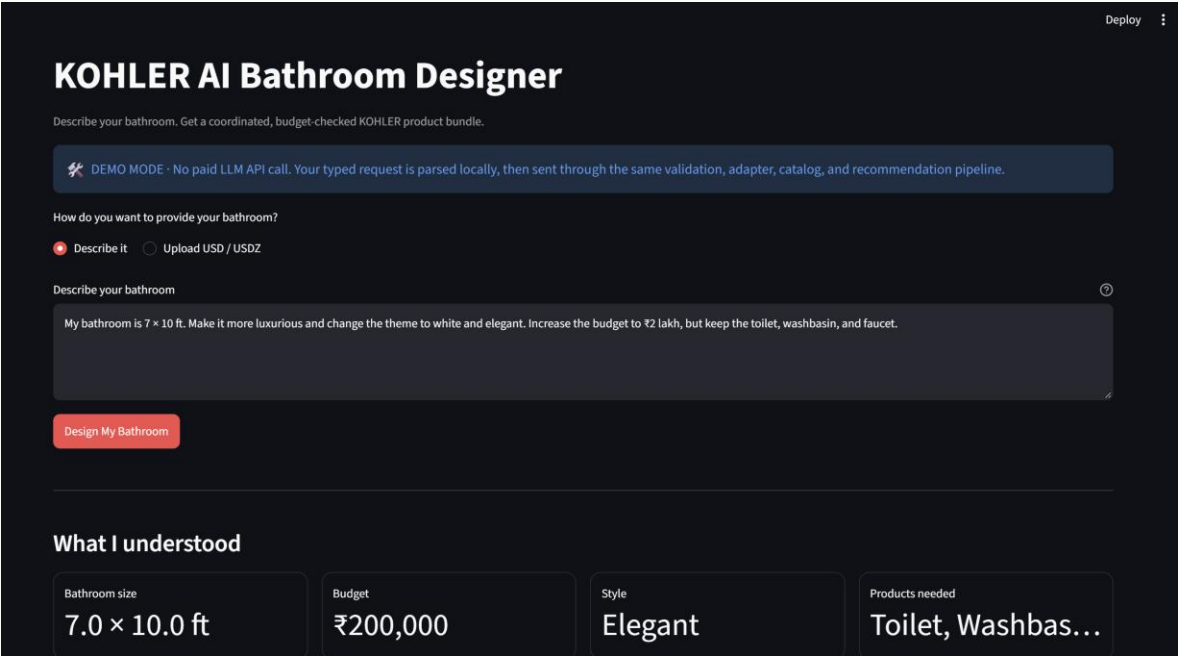
System Architecture & Workflow



The pipeline separates parsing, validation, and catalog-driven recommendation to ensure compatibility, budget checks, and layout feasibility before presenting bundles and visualizations.

Core technologies

- Python
- Streamlit
- OpenUSD
- Pandas
- Scikit-learn/ML components
- KOHLER product catalog API



Outcome: a production-ready, budget-aware design assistant that shortens the path from architectural intent to KOHLER product bundles and installable layouts.

From Requirements to Bathroom Design

We demonstrate four tightly connected feature pillars that convert intent into actionable designs and product bundles. Screenshots (placeholders): recommendation cards, USD architecture screen, conceptual layout, and product-selection cards.

① AI Requirement Understanding

Free-text parsing extracts dimensions (e.g., 10×7 ft), budget (₹1.5 lakh), style (modern), and required fixtures — producing a structured requirement manifest for validation.

② USD/USDZ Spatial Analysis

Model import extracts room envelope, openings, fixed plumbing locations, and large objects. Output: measured room geometry + object layers. Users pick which existing objects to replace or keep.

③ Intelligent Recommendations

The recommendation engine filters KOHLER catalog by compatibility, style, and budget, then ranks bundles by fit, cost, and installation constraints. Presents alternate bundles and single-item swaps.

④ What-If Redesign

Interactive scenario builder: change budget, swap fixtures, or nudge layout constraints. System revalidates compatibility and updates recommendation cards and layout preview in real time.

Impact & Future Scope



Faster Planning

Automates requirement processing and compatibility checks, reducing design iteration time.



Catalog-Driven Recommendations

Real KOHLER products drive bundles, ensuring availability, compatibility, and clearer procurement.



Architecture-Aware Design

USD/USDZ inputs produce context-aware layouts that respect existing geometry and constraints.